

1	Logarithmic functions and indices	Notes
	The functions a^x and $\log_b x$ (where b is a natural number greater than one). Use and properties of indices and logarithms including change of base. Simple manipulation of surds. Rationalising the denominator where the denominator is a pure surd.	A knowledge of the shape of the graphs of a^x and $\log_b x$ is expected, but not a formal expression for the gradient. To include: $\log_a xy = \log_a x + \log_a y,$ $\log_a \frac{x}{y} = \log_a x - \log_a y,$ $\log_a x^k = k \log_a x,$ $\log_a \frac{1}{x} = -\log_a x,$ $\log_a a = 1$ $\log_a 1 = 0$ The solution of equations of the form $a^x = b$. Students may use the change of base formulae: $\log_a x = \frac{\log_b x}{\log_b a}$ $\log_a b = \frac{1}{\log_b a}$ Students should understand what surds represent and their use for exact answers. Manipulation will be very simple. For example: $5\sqrt{3} + 2\sqrt{3} = 7\sqrt{3}$ $\sqrt{48} = 4\sqrt{3}$ $10 \times \frac{1}{\sqrt{5}} = 2\sqrt{5}$